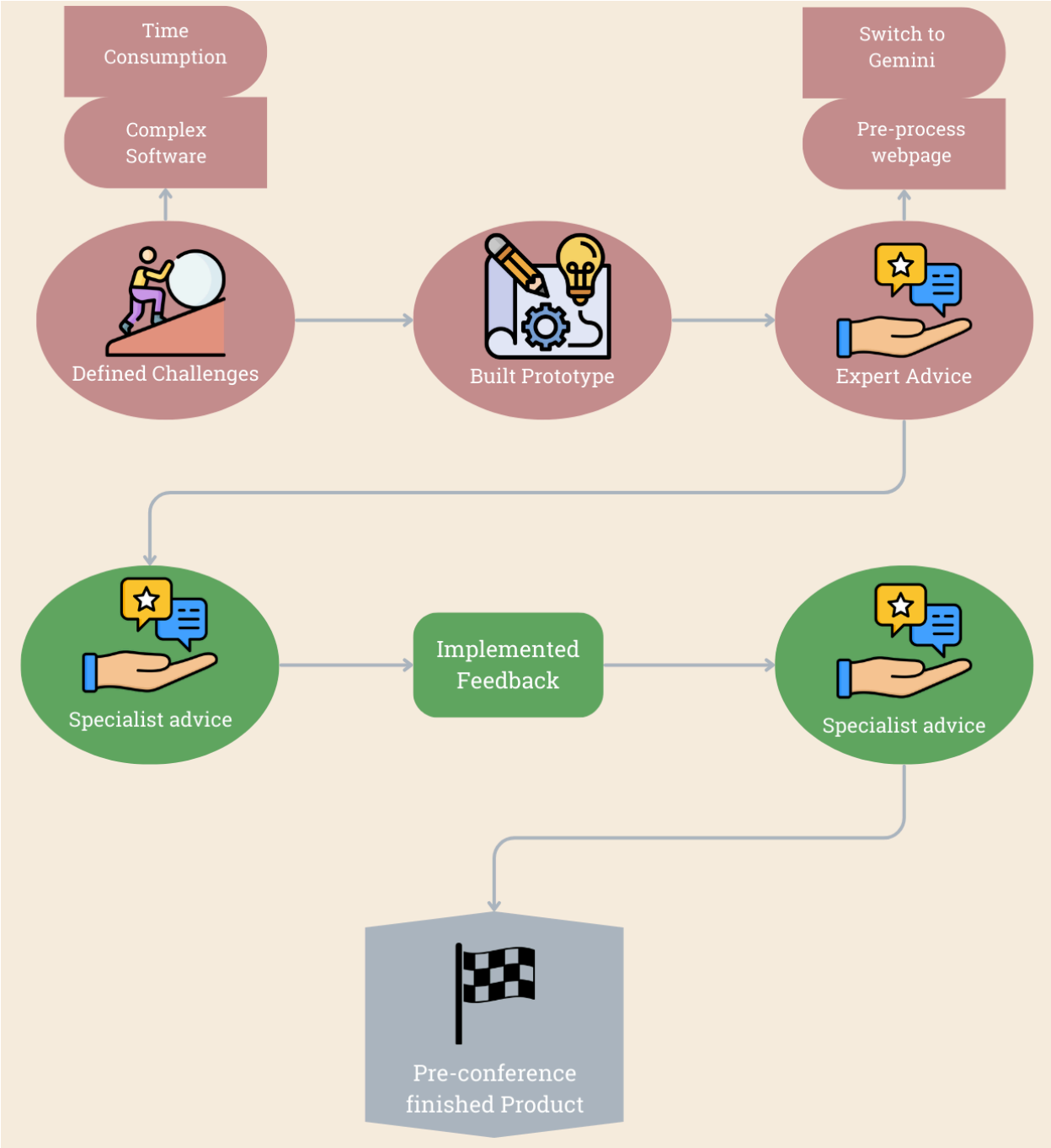
Design Documentation

Overview:



October 23rd - November 12th, 2025

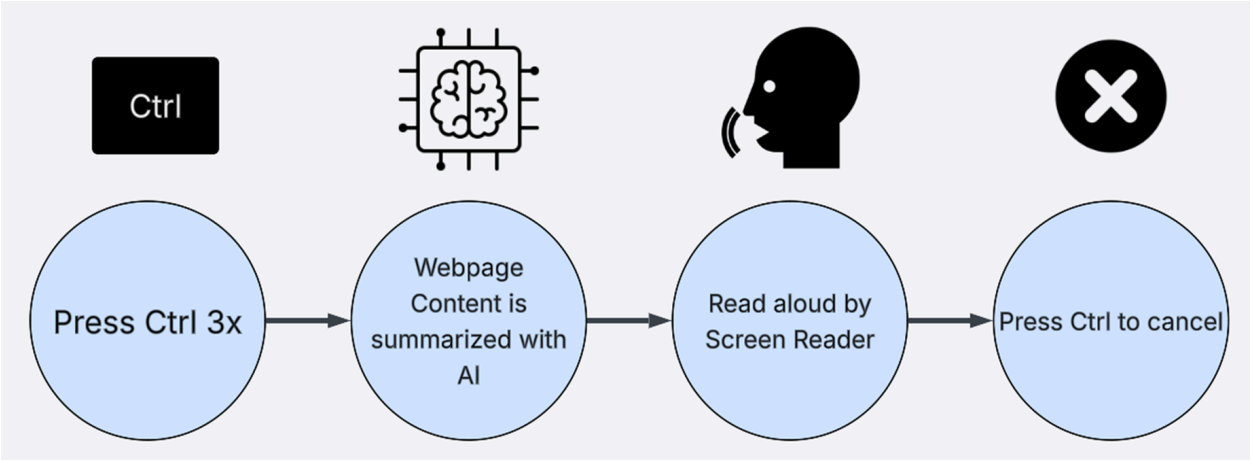
Interview

An interview between two vision specialists was conducted to define the biggest problems visually impaired users have with technology

- Compatibility issues:
    - Some screen reader software was developed in the early 2000s and isn't compatible with the latest versions of systems.
- Text-to-speech
    - This software may not correctly read what's on the screen.
    - It's also not pleasant to listen to its robotic-sounding voice
- Time consumption
    - It takes a long time to listen to content being read aloud compared to reading it with your eyes
- Simplicity
    - Screen reader software is very complex and hard to learn. This leads visually impaired users to be frustrated and neglect learning the software.

Prototype

A prototype was created with the following functionality.



The prototype can summarize a webpage's content and read it aloud. This design was chosen because:

- Summarizing a webpage's content saves them time
- **Ctrl** is very easy to remember and located on an easy-to-find spot on your keyboard

November 12th, 2025 - December 5th, 2025

Rosie, the AI Agent web assistant (version 2.0)

Rosie is an AI Agent because it helps you perform tasks. Rosie can answer your questions and perform tasks in your browser, such as opening a website.

This helps address the problems defined because:

- The user doesn't have to navigate using keyboard commands
- The user doesn't have to listen to irrelevant information being read aloud
- It's easy to learn and gets things done quickly

Interview with a software engineer - November 26th, 2025

This software engineer reviewed our software's code and gave us advice:

- Use Gemini instead of OpenAI
- Pre-process the webpage before handing it to AI
- Create thorough documentation

Interview with a previous vision specialist

A vision specialist, one that was previously interviewed, came back to test our application. The specialist gave us the following advice.

- Add sound cues during summaries and the AI Agent
- Ability to pause the screen reader
- Braille instructions for Braille display or Printer embosser

December 5th - February 28th, 2026

Version 3.0

The extension has the following capabilities:

- Summary customization
- Extension can be activated or deactivated
- AI Agent with the ability to
    - Open a website in the current tab or a new one
    - Help compose an email
    - Click buttons or links on the screen
    - Make queries on Google or YouTube
- An instructions page with printable braille instructions

All of these features can be accessed using key commands.

Interview with a vision specialist to test the application

Once all feedback from the vision specialist was incorporated, the specialist came back and tested the application again to give feedback.

- Send instructions to the Braille display
- English-to-Spanish translation
- Ability to set alarms
- Remove the need to ask for tab interaction

Version 4.0

Only some of the requested features were incorporated before the state conference. Spanish translation and Gemini were the chosen pieces of feedback. A software expert had recommended that Briefly use Gemini instead of OpenAI. The server was upgraded so it could translate English into Spanish and have the ability to make API calls to either Gemini or OpenAI.

Conclusion

The software assists visually impaired users on the internet by saving them time. The three main challenges defined from the interview in October were:

- Time consumption
- Complex screen reader software
- “Robotic” sounding voices

The software was designed to solve these problems by:

- Summarizing webpages
    - Reduces the amount of time spent listening
- Easy-to-remember and locate key commands
    - Makes it very easy to learn
- Rosie, the AI Agent
    - Saves time in performing tasks on the internet